



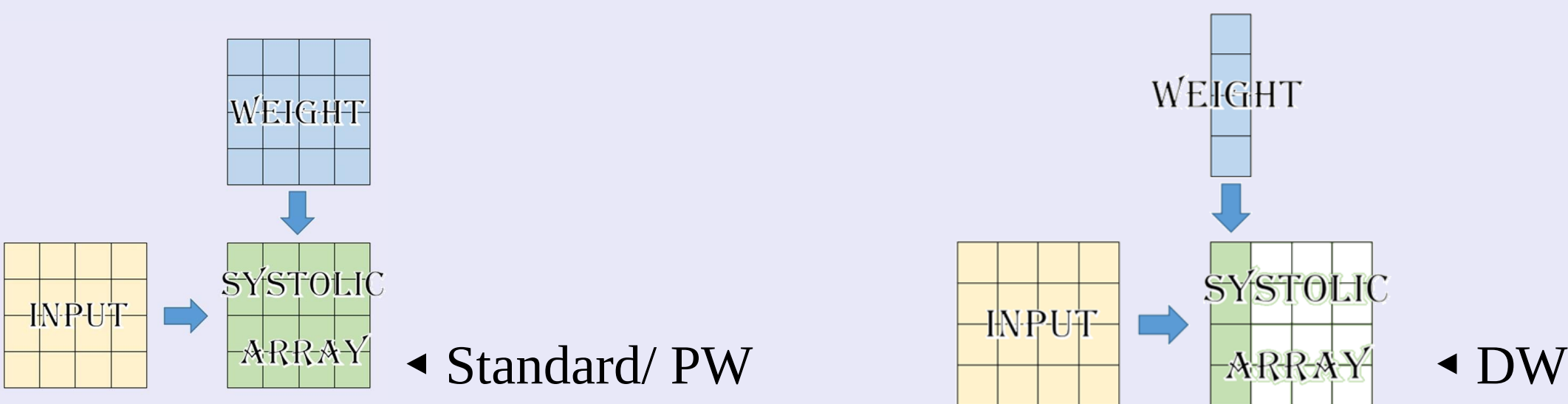
# Configurable DNN Accelerator with Distributed SRAMs

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**Abstract** - Traditional AI accelerators often face low utilization rates during depthwise-separable (DW) convolution computations. To address this, we propose an architecture featuring distributed buffers instead of a global one to enhance efficiency. This poster shows the architecture and how we implemented it.

## I. Introduction

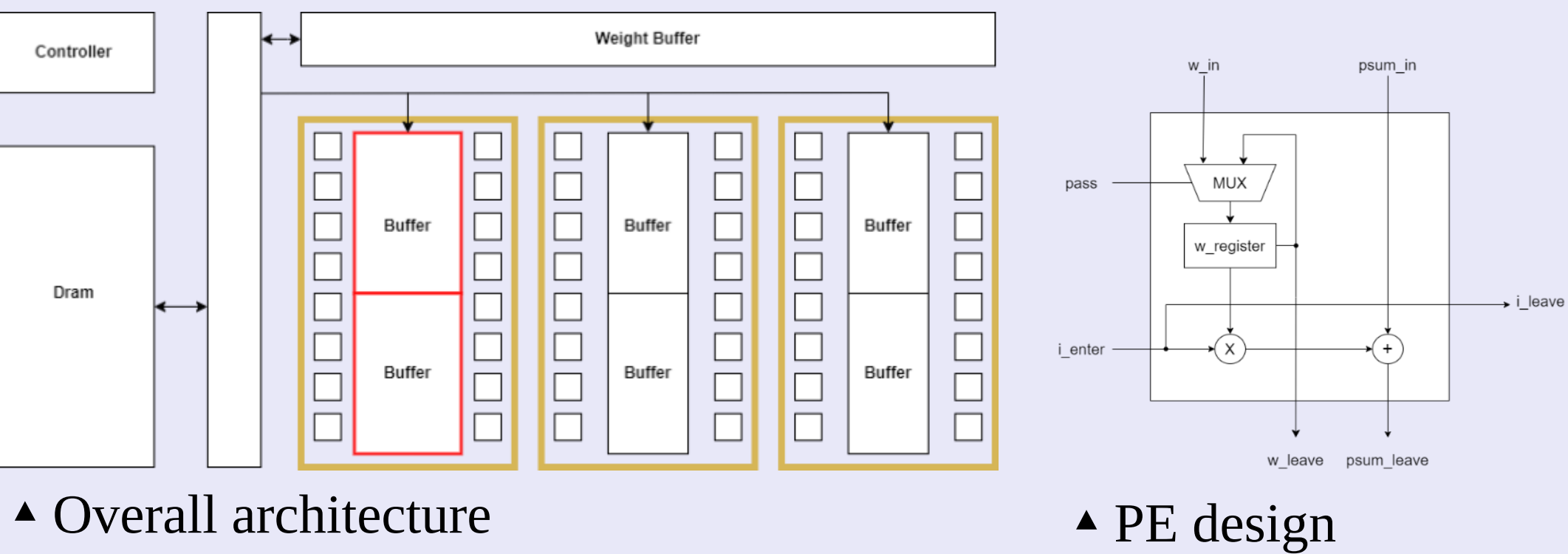
The following figures show the dataflow of different convolution computations. We can see that the utilization rate of PEs is low due to DW convolution’s inherent lack of input reuse characteristics. With our method, we can ensure the utilization rate when performing DW convolution as well as keep standard and PW convolution work functionally.



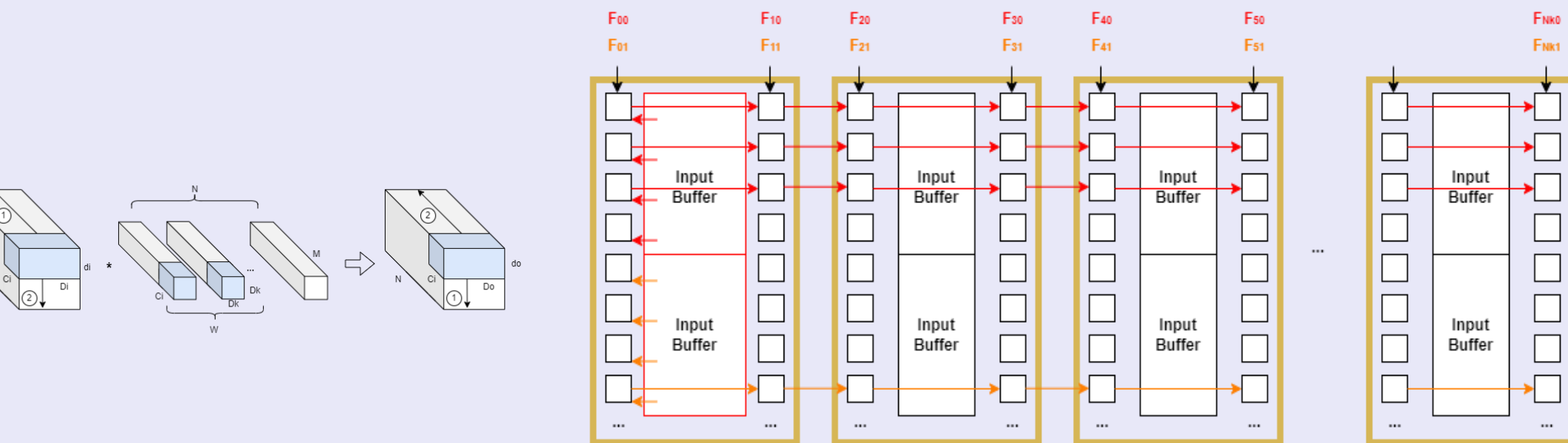
## II. Methods

Central to our architecture is the division into distinct units, each comprising an SRAM buffer flanked by two columns of PEs.

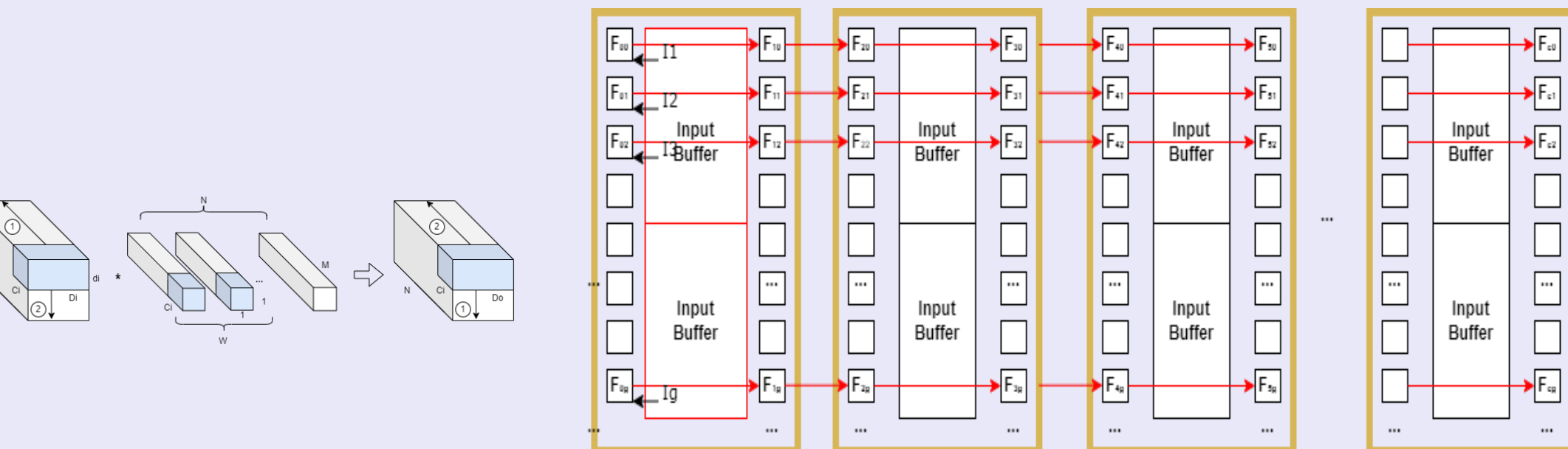
First, we need to design PE which can load the input, partial sum, and weight. The “pass” variable can ensure PE restores or passes the weight. After the calculation, PE will pass the input (in standard and PW) and the new partial sum to the next PE.



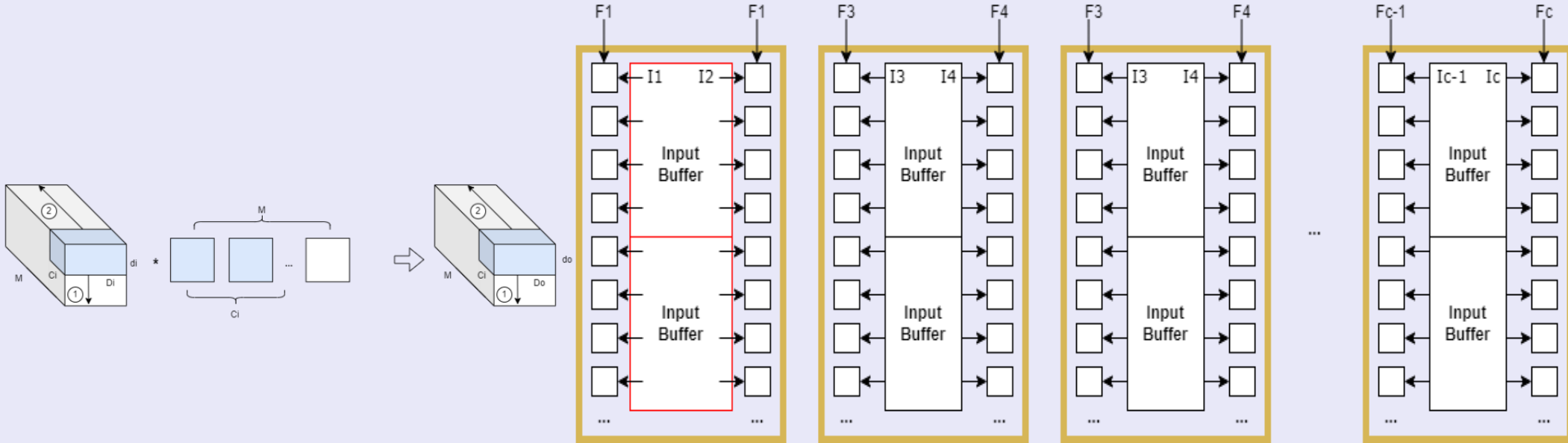
This modular arrangement ensures optimal performance across diverse convolutions. The following figures illustrate the dynamic dataflows within our architecture across three prevalent convolution types.



▲ Dataflow and tiling of standard convolution



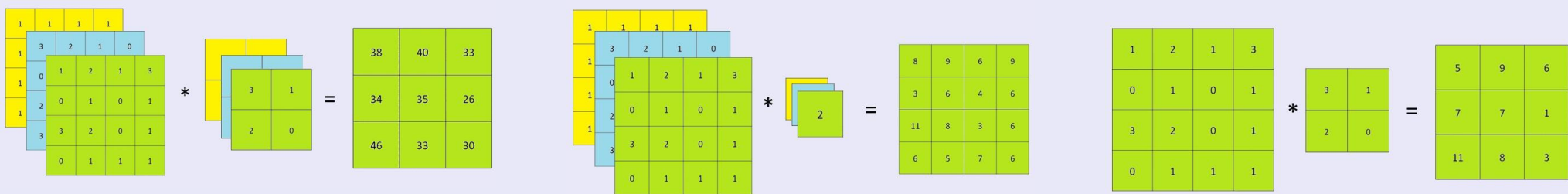
▲ Dataflow and tiling of PW convolution



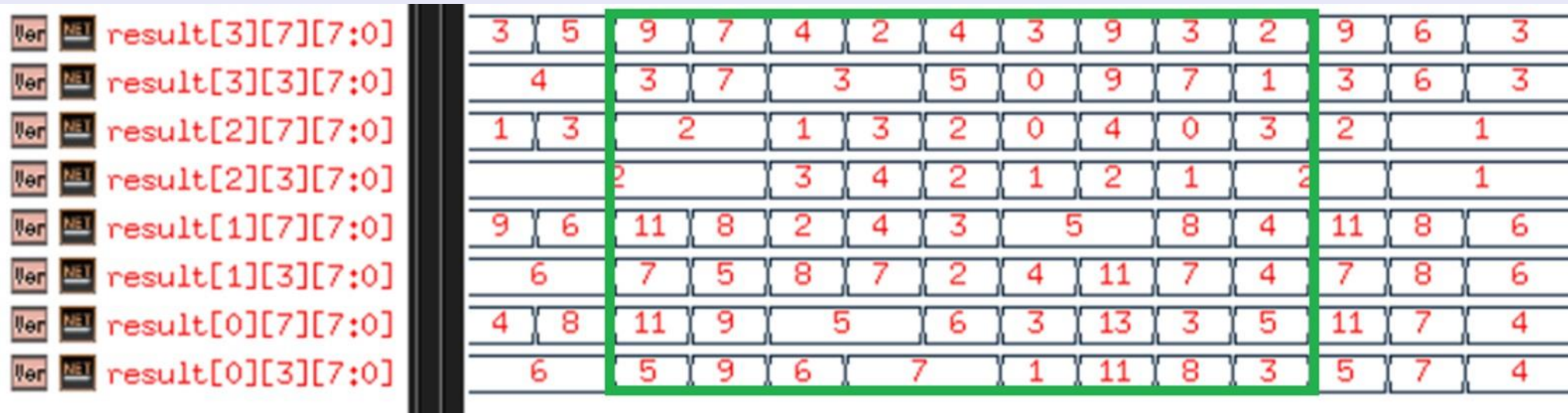
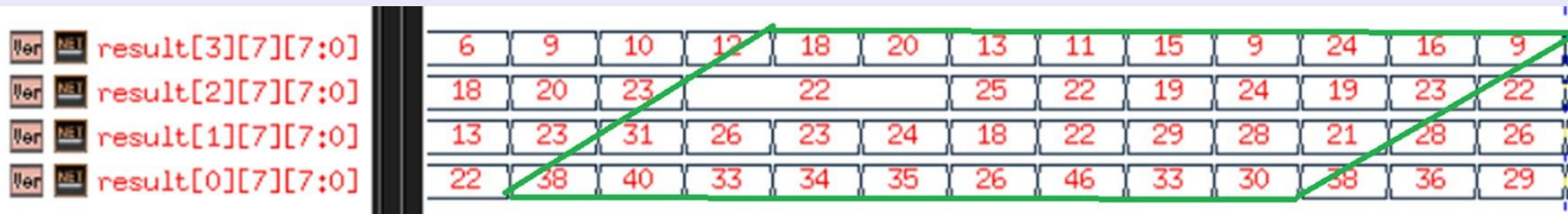
▲ Dataflow and tiling of DW convolution

## III. Results

### 1. Waveform



▲ Input maps, kernel maps, and the expected output maps (from left to right: standard, PW, and DW)



▲ Experimented output maps (from top to bottom: standard, PW, and DW)

### 2. Area and Timing Report

| Item                | Number | Item             | Area          |
|---------------------|--------|------------------|---------------|
| Ports               | 3653   | Combinational    | 11873.837060  |
| Nets                | 6852   | Buf/Inv          | 445.233615    |
| Cells               | 4097   | Noncombinational | 19559.232044  |
| Combinational cells | 2786   | Macros/Black Box | 706633.773438 |
| Sequential cells    | 1260   | Total cell       | 738066.842542 |
| Macros/Black boxes  | 5      |                  |               |
| Buf/Inv references  | 206    |                  |               |
|                     | 21     |                  |               |

▲ Area report

| Item               | Time(ns) |
|--------------------|----------|
| data required time | 2.85     |
| data arrival time  | -2.60    |
| slack (MET)        | 0.25     |

▲ Timing report

## IV. Conclusion

The PE model we've developed can accurately compute Standard, Depthwise, and Pointwise operations. Moreover, it can perform secondary computations when the total number of kernels exceeds the number of PEs, and in the case of Depthwise, it can increase the utilization rate of the model.

## Reference

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